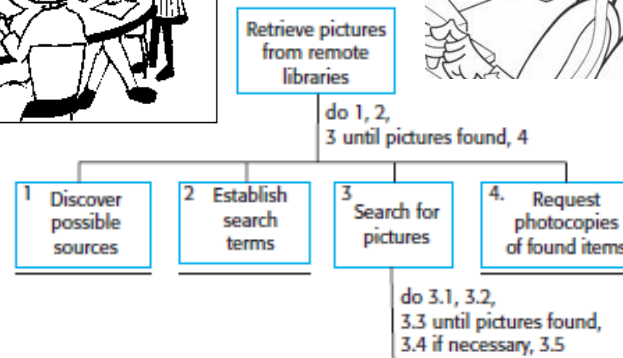
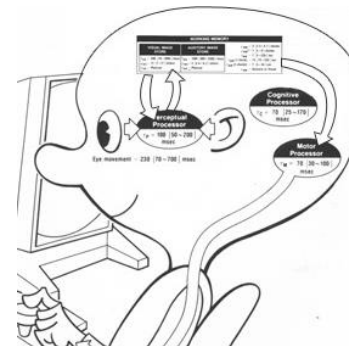




Other methods for design and evaluation



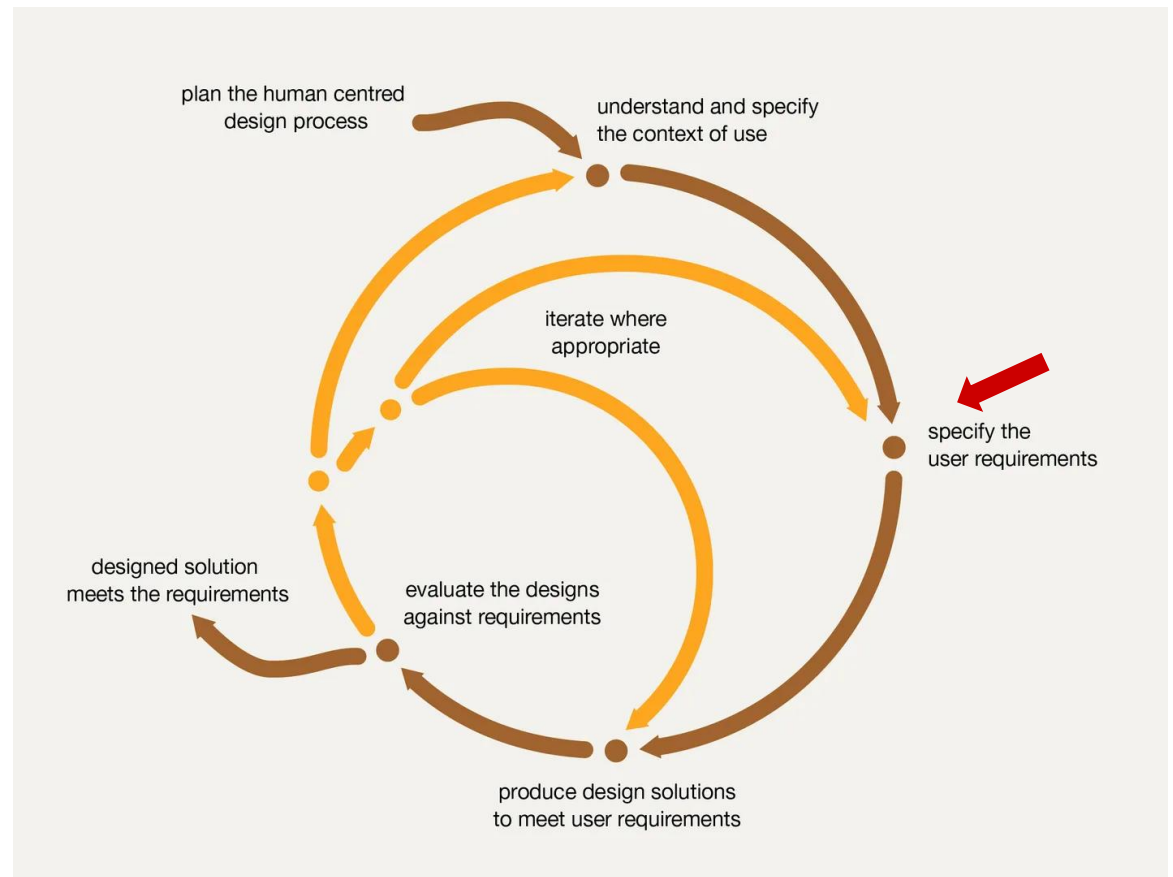
Human-Centered Design

Starts by user research

Is iterative

Includes evaluation in each cycle

We need models ...



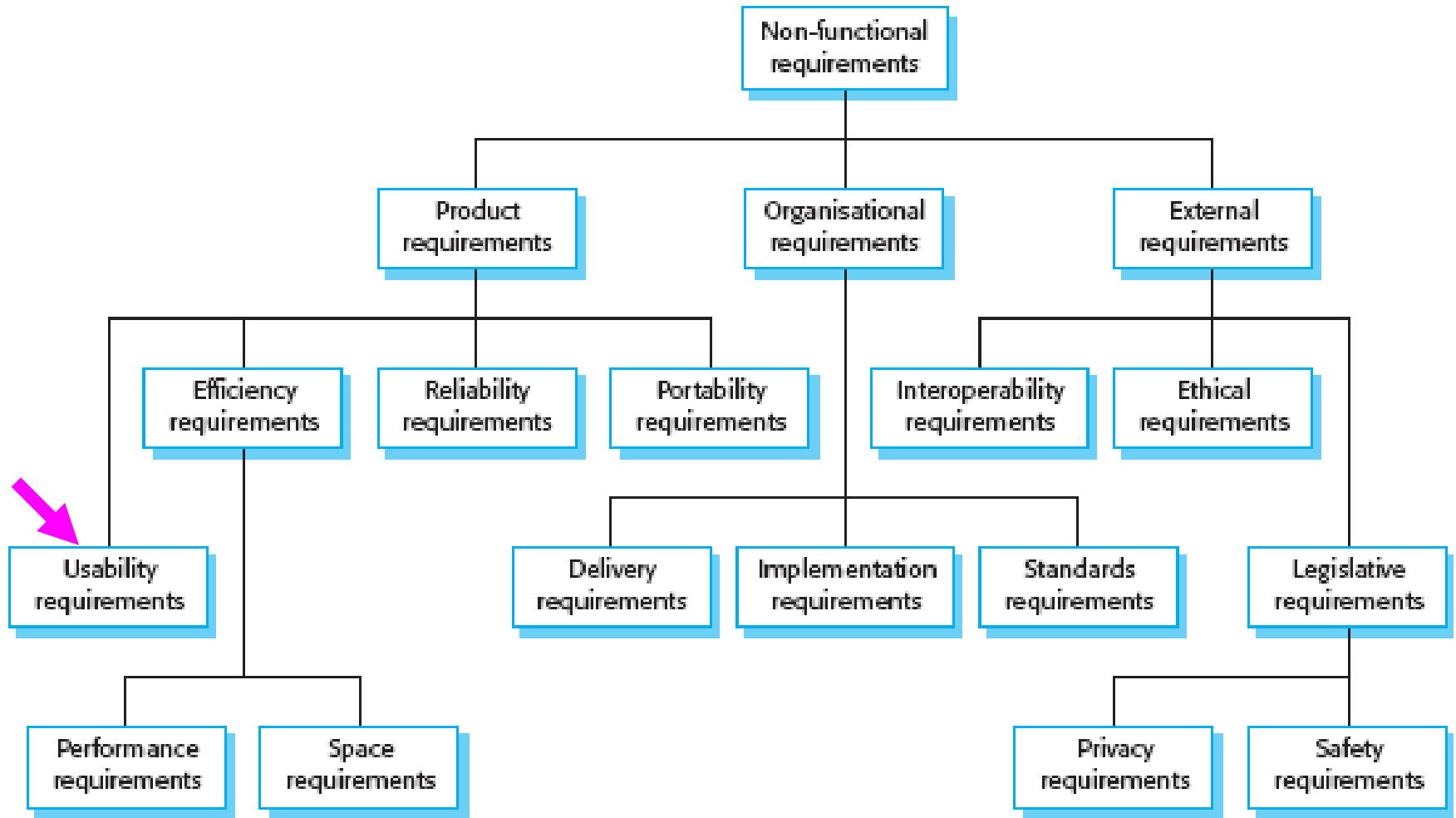
[What is The Interactive Design Process? | IxDF](#)

- All engineering fields use models:
 - To prescribe – directly contribute to the design
 - To evaluate – does the design have the needed characteristics?
- Models are needed also in Interactive systems design (you already know and have applied a few...)
- Several types of models may be used throughout the design of interactive systems and user interfaces:
 - User models
 - Task analysis
 - Dialog notation
 - ...

Models to obtain user requirements

- Obtaining user requirements is very important in S/W engineering
- But often focuses on **functional requirements**: what the system must do
- Overlooking **non-functional requirements**, as if it is:
 - acceptable
 - usable
- There are several models to capture user requirements

Recap: Non-functional requirements



(Sommerville, 2016)

Models to obtain user requirements

- *Socio-technical models* - acknowledge that technology is used within an organization
- *Soft System methodologies*- put the emphasis on the understanding of the situation and not on getting a solution
- *Participatory Design* – encompasses the complete development cycle and includes users as active project team members, not only as evaluation participants

Participatory design

- Users are involved as domain experts (e.g. business representatives and users) along the complete process and work together with developers to design a solution
 - It is work oriented and not system oriented
 - It is collaborative- users contribute to all phases
 - It is iterative – design is evaluated and reviewed in every phase

Participatory design

- Uses a set of techniques (that can be used in other contexts) to help transfer information from users to designers:
 - Brainstorming
 - Scenarios
 - Story boarding
 - Workshops
 - Paper and pencil exercises ...



[Usability First - Usability Glossary - Participatory Design | Usability First](#)

Task analysis

- What it is and how it can be performed (more or less formally)
(Note that in Lab classes we started by doing it in an informal way...)
- Techniques
- Sources of information



[Hierarchical Task Analysis | Usability Body of Knowledge](#)
[Hierarchical Task Analysis | Digital Healthcare Research](#)

Task analysis

- It is the analysis of how people perform their work
 - what they do
 - what they use
 - what they need to know
- Daily-life example: vacuum cleaning a house
 - Get the vacuum cleaner
 - Choose the adequate attachment
 - Clean the rooms
 - Empty the bag when it is full
 - Put the vacuum cleaner and attachments away

Users have to know about vacuum cleaners, rooms, ...

Task Analysis

- Observation (of various types) is a fundamental tool
- It can be used to:
 - produce documentation and training materials
(the observation of how existing systems are used is enough)
 - design new systems
(work-re-engineering is usually necessary)

Task Analysis approaches

- There are a lot of approaches and methods; we have seen first an informal way, but there are more formal methods
 - Task decomposition ✓
 - Knowledge based
 - Relation and entities based

- more formal methods:
 - Task decomposition - divides tasks into subtasks that must be performed in a specific sequence
 - Knowledge based – considers what users need to know about the objects and actions involved in performing the task and how knowledge is organized
 - Relation based - is focused on actors and objects, relations among them and the actions they perform

Task Decomposition

- Hierarchical Task Analysis (HTA) is one of the most used task analysis techniques and produces:
 - a task and sub-task hierarchy
 - plans with a sequence and execution conditions

Simple daily life example: vacuum cleaning the house:

- 0. in order to clean the house
 - 1. get the vacuum cleaner
 - 2. fix the appropriate attachment
 - 3. clean the rooms
 - 3.1. clean the hall
 - 3.2. clean the bedrooms
 - 4. empty the dust bag
 - 5. put the vacuum cleaner and attachments away

Plan 0: do 1 – 2 – 3 – 5 in that order
when the dust bag gets full do 4

Plan 3: do any of 3.1, 3.2, or 3.3 in any order depending on which rooms need cleaning

Plan 3 could be more specific; what if it were varnishing the house?

- Where should the decomposition stop?
- The decomposition detail depends on the goals of each task analysis
- Example: in a factory what should be done in an emergency
 0. in a emergency
 1. Read the alarms
 2. Determine the corrective actions
 3. Execute the correction actions
- If the goal is
 - installing a monitoring system → expand 1 e 3
 - produce operation manuals → expand 2

- Where should the decomposition stop?
- A stop decomposition rule:

Stop if $P \times C < \text{a specific value}$

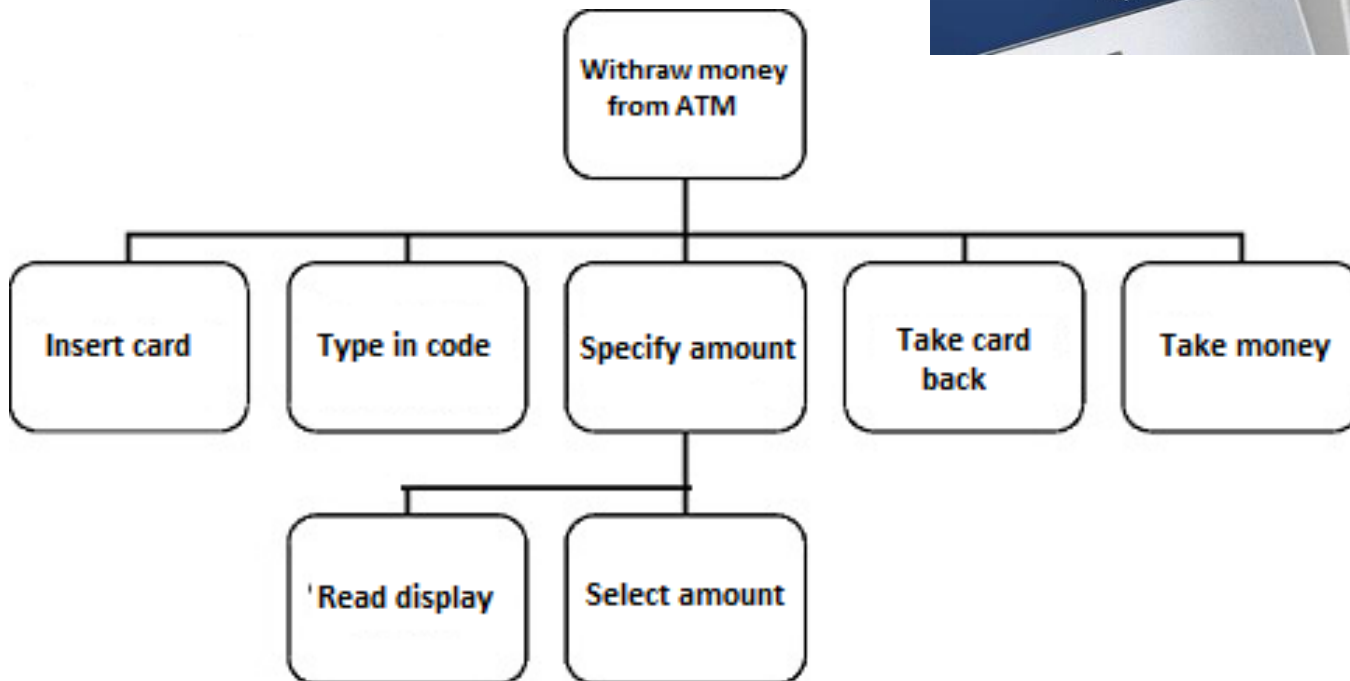
error cost

Probability of making an error

i.e.: simple tasks do not need decomposition unless they are critical!

Incomplete decomposition of the task: Withdraw money from an ATM

Can you fix and complete it?



Another daily life example:

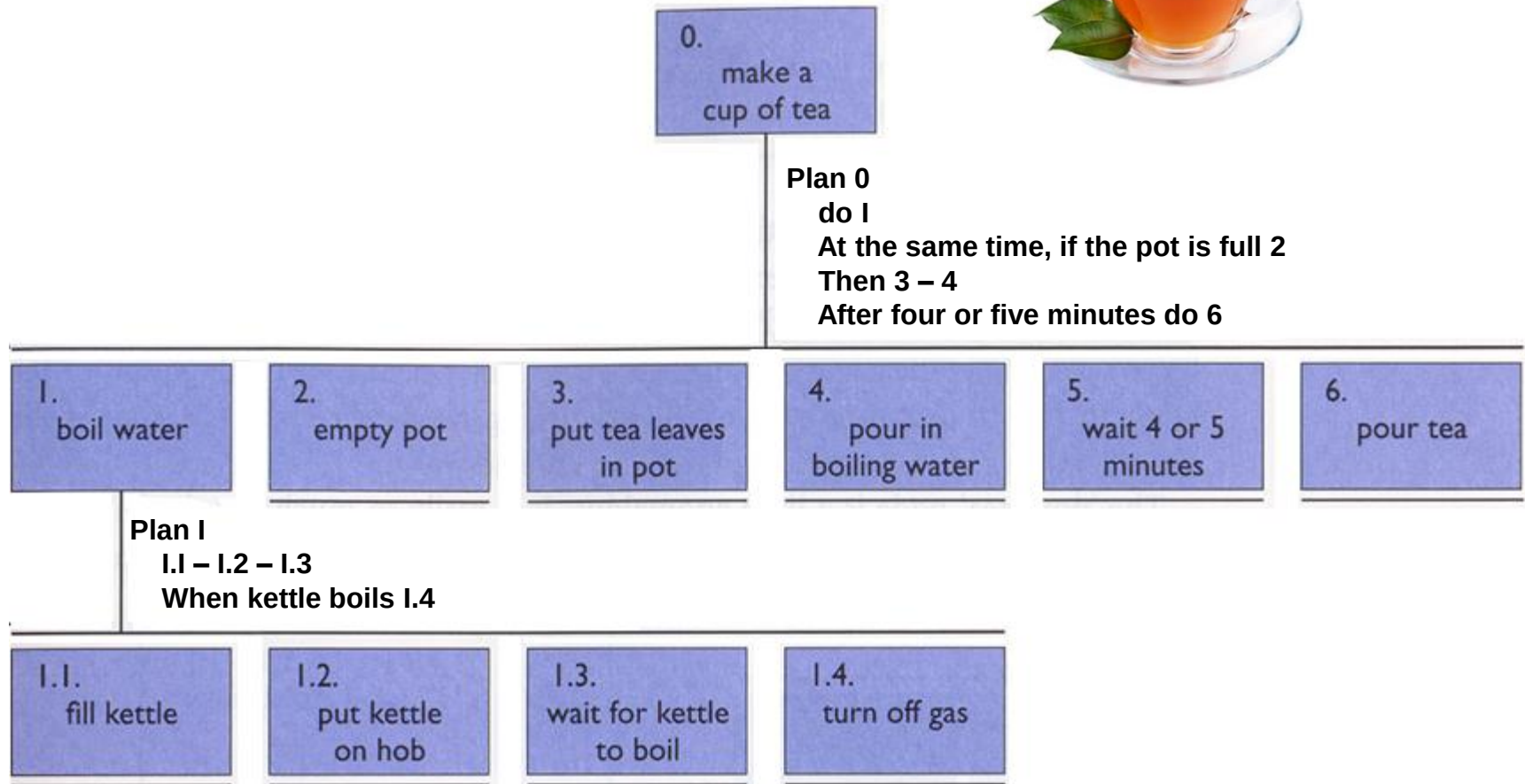
Preparing a cup of tea

Can you do a HTA describing this task?

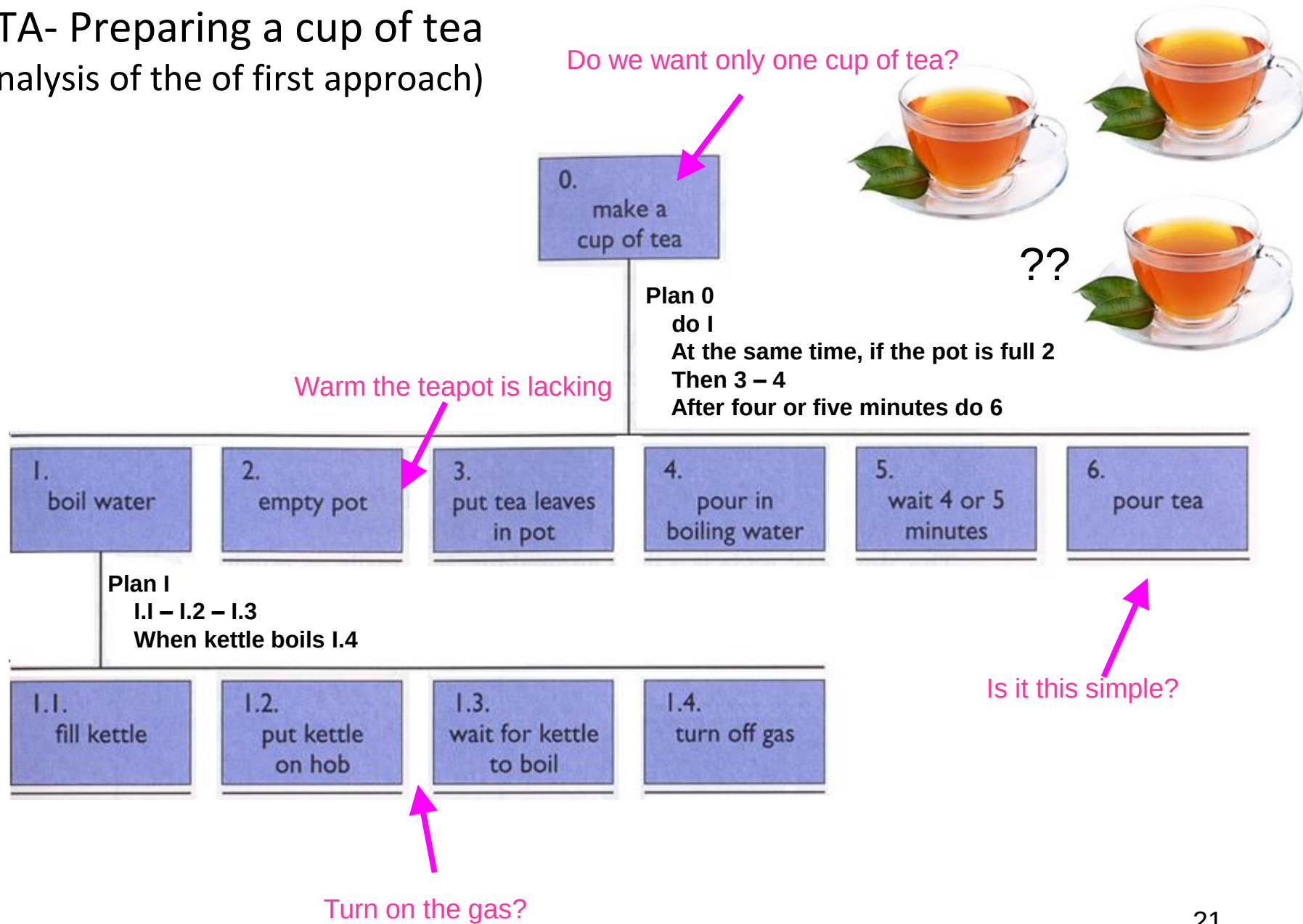


HTA- Preparing a cup of tea

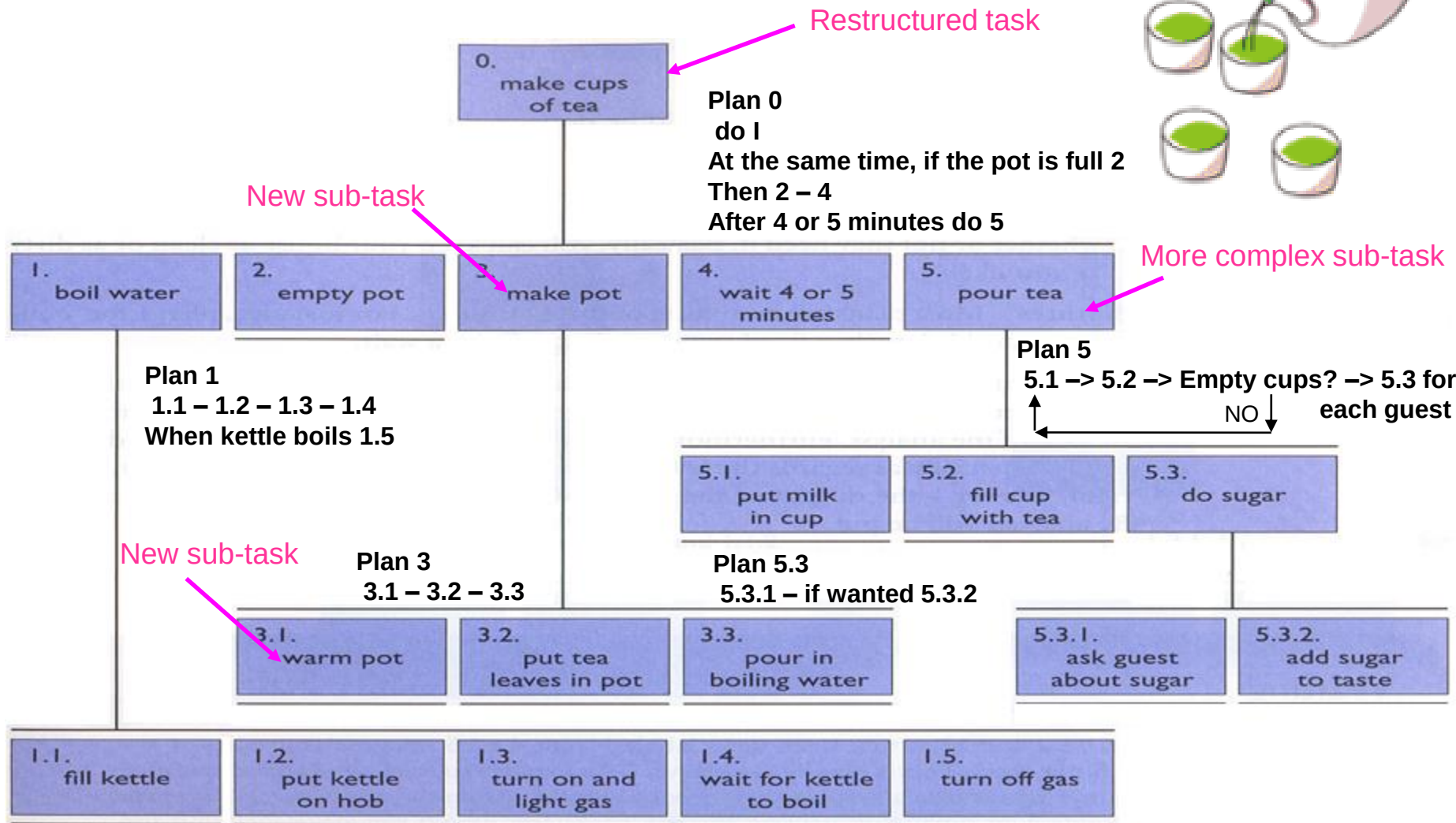
(graphical representation
of first approach)



HTA- Preparing a cup of tea (analysis of the of first approach)



HTA- Prepare several cups of tea (a new hierarchy)



Plan types in HTA

- Fixed sequence (plan 3 – prepare the teapot)
- Optional tasks (5.3 sugar?)
- Waiting for events (4- wait 4 or 5 minutes)
- Cycles (plan 5 – serve tea)
- Time sharing (1 and 2 prepare teapot, boil water)
- Random (vacuum cleaning rooms)
- Mix of several types

- The result of the analysis depends a lot on the experience of the analyst
- **Different analysts usually produce different results** (mainly at the detail level) varying with the goal of the analyst

Task analysis information sources

- The quality of task analysis results cannot be better than the original data
“garbage in garbage out”
- The process of analysis in general triggers new questions, thus several phases of data collection and analysis are needed
- There are several types of information sources:
 - Documentation
 - Observation → (expensive)
 - Interviews →



Documentation:

- Manuals, instruction books, training documentation ... are very good information sources
- But they describe what people are supposed to do, not what they actually do
- System's manuals usually describe functionality, not how they are used
- Observation and user interviews should be performed based on this information
- Be careful with user interviews!!

Interviews:

- Interviewing domain experts is a good way of getting information about the task; should include:
 - General questions (e.g. a typical day)
 - Specific questions (e.g. why did you do that?)
 - Task decomposition (~ HTA)

Observation:

- It is always necessary to perform (formal or informal) observation to understand the tasks
- Reading documentation and observing users is a good starting point
- More observation should follow:
 - In the lab
 - In the field
 - Passive (only observation)
 - Active (questions, post-task walkthrough)

Using Task Analysis:

- May be used in:
 - Manuals and teaching materials
 - High-level system design
 - Detailed design of the system user interface
- In the first case users are observed while performing tasks using the system
- In the other cases task analysis contributes to the design of the new system

Manuals and teaching materials

- The first task analysis techniques were developed to train people to perform a task

(e.g.: clean a gun)

- HTA may be used to structure manuals or tutorials

(e.g: how to prepare tea)

- May be used also to help users transfer from one system to another

Obtaining requirements

- TA is not a direct way of obtaining a system requirements as it refers to an already existing system and not to the new one
- However, it may give a good contribution
- In general the new system includes old and new functionality
- TA may help determine:
 - What objects, tasks, etc., should be maintained
 - What should be included

II- User Models

- User models of the users' mental, perceptual and motor processes (GOMS, KLM...)
- Personas- fictional characters based upon research in order to represent the different types of users
- ...

[Personas | The Encyclopedia of Human-Computer Interaction, 2nd Ed.](#)

GOMS- Goals, Operators, Methods and Selections

- Proposed by Card, Moran and Newell, 1983
- A GOMS decomposition has the following elements:
- **Goals:** what the user wants to attain
- **Operators:** basic operations that the user has to perform to use the system; may affect the system or not (press a key or read a message)
- **Methods:** possible decompositions of the goal into sub-goals (e.g. Select an option “Save” or press “ctrl S”)
- **Selections:** rules to select the possible methods (taking into account the type of user and the system status)

[Goals, Operators, Methods, and Selection Rules | Digital Healthcare Research](#)

Example: 'save' a file: using two common ways

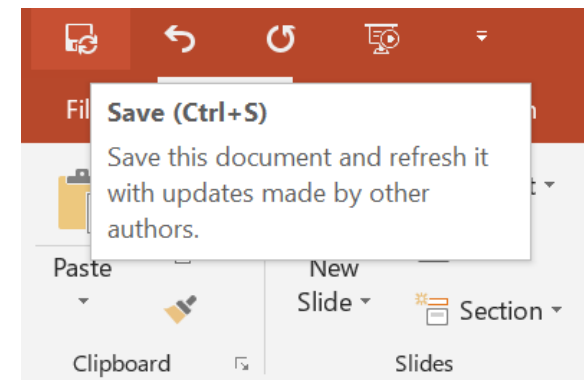
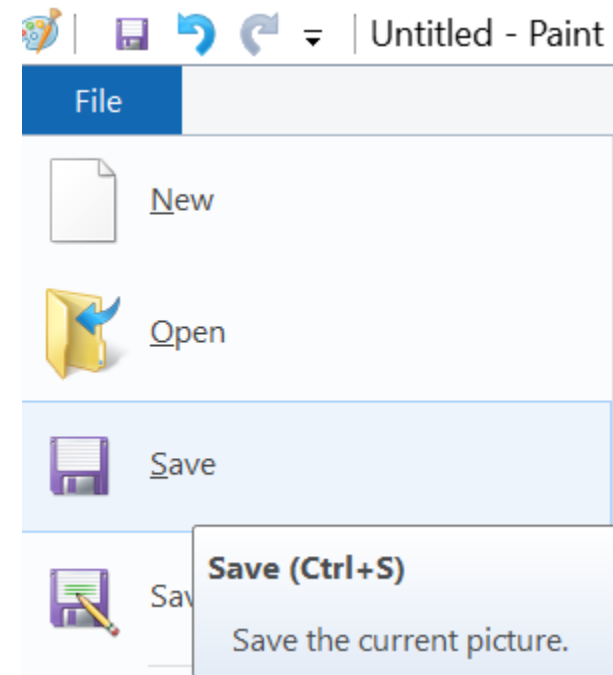
- **GOAL: SAVE-A-DOCUMENT**

- . [select GOAL: USE-SAVE-OPTION-METHOD
 - . MOVE-POINTER-TO-MENU-BAR
 - . CLICK-OVER-FILE-MENU
 - . MOVE-POINTER-TO-SAVE-ICON
 - . CLICK-SAVE-ICON
- . **GOAL: USE-CTRLS-METHOD**
 - . PRESS-'CTRL'+ 'S'-KEYS

User BSS:

Rule 1: **USE-CTRLS-METHOD** unless other rule applies

Rule 2: If has hand on mouse/touch pad **USE-SAVE-OPTION-METHOD**



GOMS- Goals, Operators, Methods and Selections

- A typical GOMS analysis consists in decomposing a high level goal in a sequence sub-goals
- Selection rules must be adjusted to the user profile
- Analyzing the structure of the GOMS decomposition may give an **approximate measure** of :
 - Short Term Memory load (depth of the goal structure)
 - Time needed (a time for each operator)

Example: Copy/scan an article from a journal

- Goal: Photocopy/scan-paper
- . Goal: Locate-article
- Goal: Photocopy/scan-page repeat
- Goal: Orient-page
- open cover
- select-page
- position-paper
- close-cover
- Goal: Press-copy-button
- Goal: Verify-copy (only if copy)
- locate-out-tray
- examine-copy
- Goal: Collect/send-copy
- locate-out-tray
- remove-copy/send pdf (outer goal satisfied)
- Goal: Retrieve-journal
- open-cover
- remove-journal
- close-cover

Closure problem

(the user attains the goal before the task is complete)

The “closure problem”



In earlier ATMs the money was given before returning the card

... many users left the card:
their goal was getting money!

This was changed.

The copies usually are available to the user before they remove the original from the photocopier and walk away!

To prevent this, the overall goal should be satisfied only after removing the original



The “closure problem” in MultiBanco



In stores usually these are the following steps:

- Insert the card
- Insert the pin code
- Transaction approval -> audio signal
- Remove the card
- Receipt is handed to the client

At the ATMs the money is given (goal satisfaction) only after the card is removed by the client



These procedures help not to forget the card!

GOMS- Goals, Operators, Methods and Selections

- Capacities:
 - It has been used in cognitive model research
 - It may describe adequately how **experienced users** perform **routine tasks**
 - Associated to a device model allows time estimates

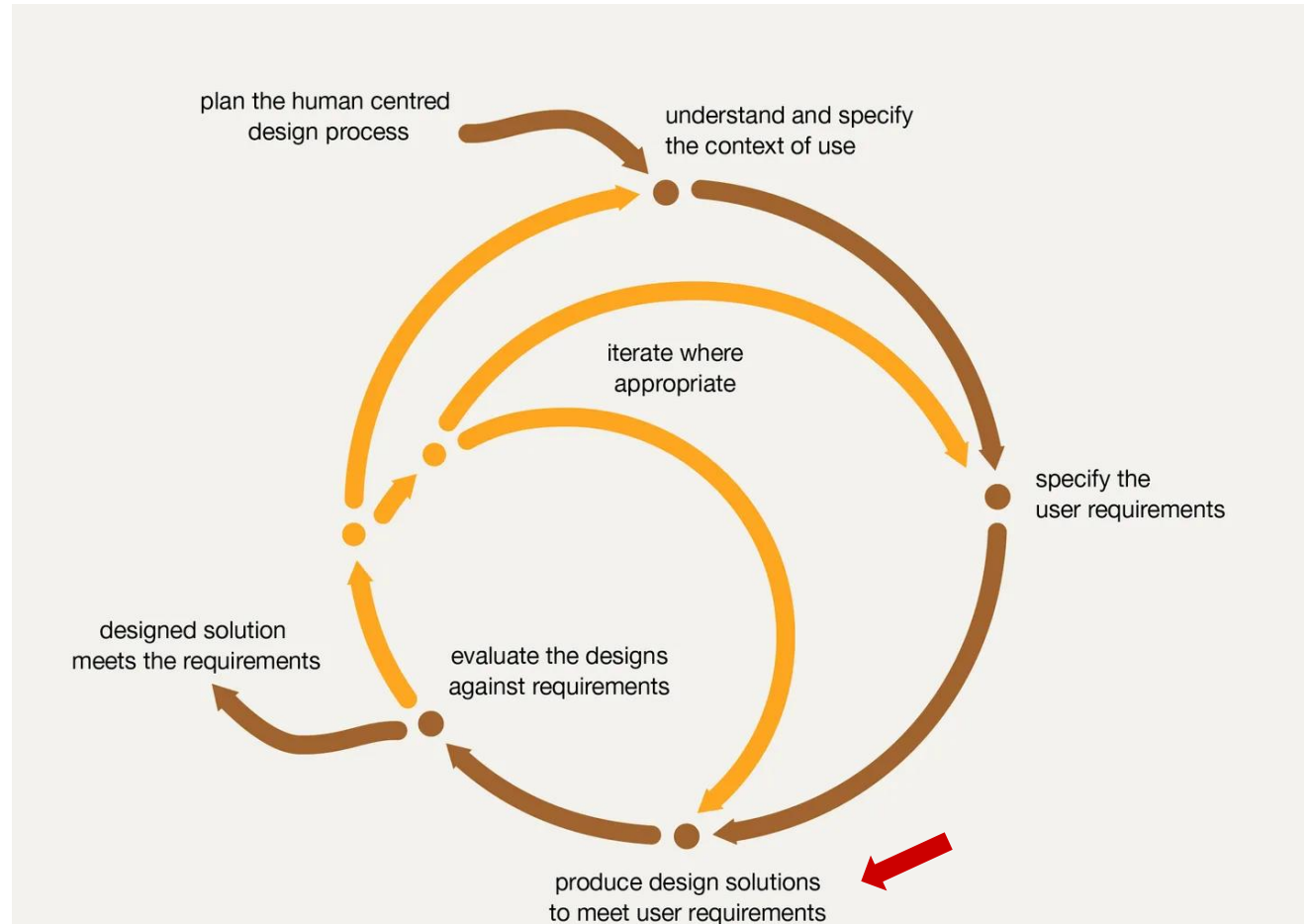
Limitations:

- It does not give information concerning user knowledge to estimate training or transfer times
- It requires experience to apply correctly

Differences between GOMs and Task Analysis (TA)

- The scope of Task Analysis is very wide
- TA models also aspects of the real world not part of the system
(example: feeding paper into a printer; getting paper documents)
- TA describes the tasks users perform from an external point of view and has more detail
- GOMS aims at understanding the user's cognitive processes while performing the task
- TA is more used in early phases of the S/W lifecycle and GOMS for evaluation

What next?

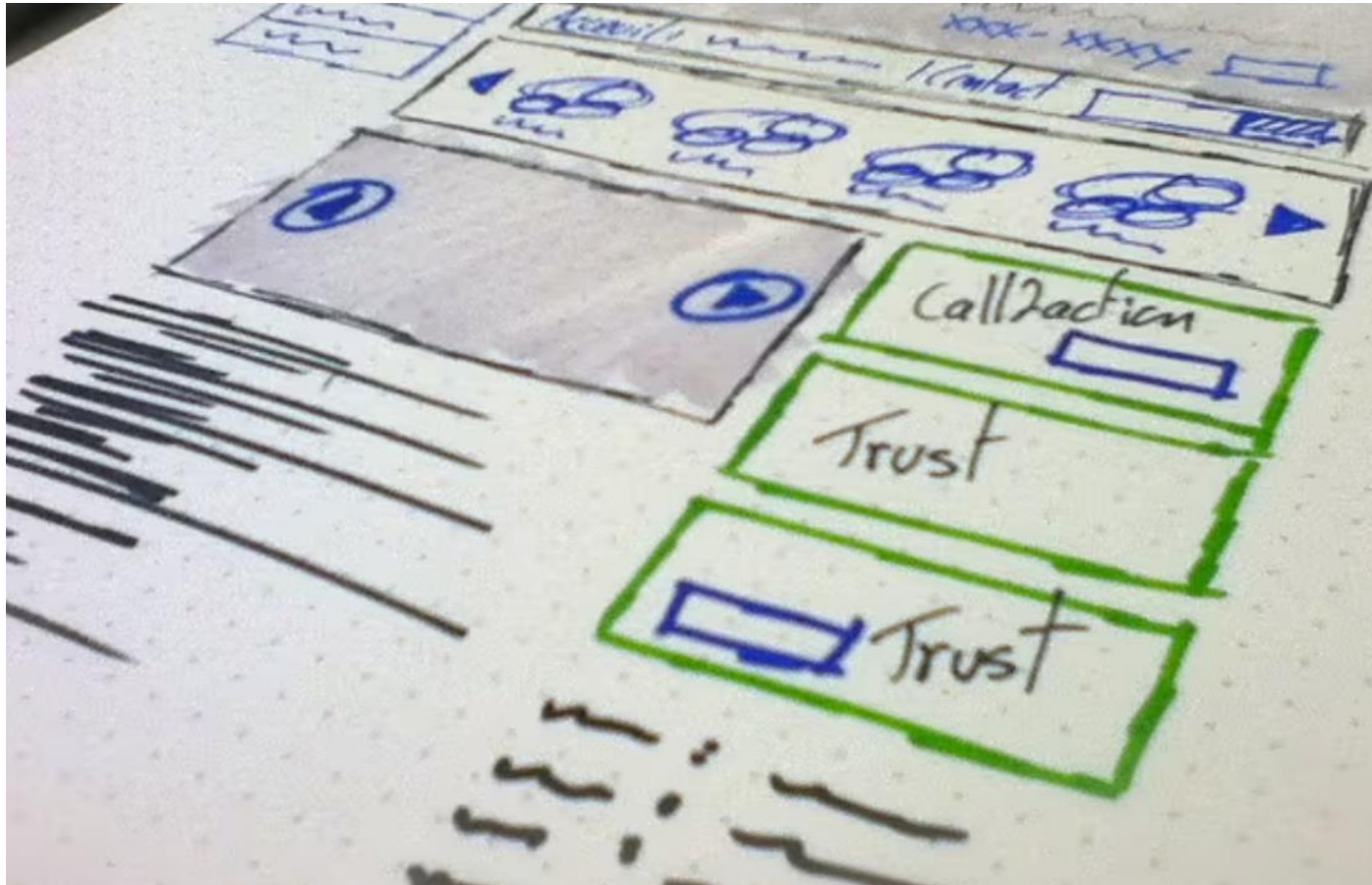


[What is The Interactive Design Process? | IxDF](#)

How can we move from the requirements and start designing the interface?

We prototype!

Prototyping



[Rapid Prototyping, Faking It Until You Make it in a UX Driven World | IxDF](#)

Main bibliography

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- Interaction Design Foundation – IxDF, *How to improve your UX designs with Task Analysis*. Interaction Design Foundation – IxDF, 2023

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https://www.academia.edu/40407620/Designing_Interactive_Systems_A_comprehensive_guide_to_HCI_UX_and_interaction_design

- Alan Dix, J. Finley, G. Abowd, R. Beale, *Human-Computer Interaction*, 3rd ed., Prentice Hall, 2004

- Ian Sommerville, *Software Engineering*, 10th ed., Pearson, 2016

Interesting links:

- <http://www.usabilitybok.org/goms>
- <http://www.usabilitybok.org/task-analysis>
- <http://web.mit.edu/6.813/www/sp17/classes/09-more-efficiency/>

“Look inside” the following books:

Y. Rogers, H. Sharp, J. Preece, *Interaction Design: beyond human-computer interaction*, 3rd Edition, Willey, 2011,

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B. Weyers, J. Bowen, A. Dix, P. Palanque (eds), *The Handbook of Formal Methods in Human-Computer Interaction*, Springer, 2017

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